



PROGRESS REPORT

Response to Review Feedback

Smart Healthcare Platform — Iteration 2

New this iteration:

- Computed model confidence (5-factor, data-driven)
- NEWS2-aligned vital sign scoring
- AI Assistant — RAG (general) + patient-context mode
- 37-test automated suite, including a real bug caught and fixed

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Executive Summary

CHAPTER

This report documents the second iteration of the Smart Healthcare Platform, produced in direct response to the review feedback received on the first progress report. Every point raised has been addressed with either a concrete implementation, a documented and justified design decision, or an explicit "not yet addressed" flag where that is the honest status.

#	Feedback point	Status
1	Model confidence was a fixed value	DONE
2	Risk weights lacked clinical grounding	DONE
3	French/Arabic NLP depth (NER vs. keyword)	PARTIAL
4	No test suite / validation	DONE
5	Security/compliance basics (HTTPS, encryption, retention)	OPEN
6	Synchronous pipeline	PARTIAL
—	AI Assistant design note & guardrails	DONE

What changed since the last report

The risk engine now computes an explainable, data-driven confidence score instead of a fixed constant; vital-sign thresholds are aligned with NEWS2; a 37-test automated suite now covers score capping, negation edge cases, and NEWS2 boundaries — and, in the process of writing these tests, a real production bug was found and fixed (detailed in Chapter 4). A new AI Assistant feature has also been built, scoped exactly along the lines the review recommended before this feedback was even received: RAG-grounded, disclaimer-carrying, fully logged, and access-scoped by role.

Response to Feedback — Point by Point

CHAPTER

2.1 Point 1 — Model confidence

Done

The fixed value (0.6 if `factors` else 0.35) has been replaced with a computed, weighted, five-factor confidence score. See Chapter 5 for the full formula and worked examples with real output values (51%, 74%, 95%).

2.2 Point 2 — Clinical grounding of risk weights

Done

Vital-sign scoring (temperature, heart rate, oxygen saturation, systolic blood pressure) is now derived from NEWS2 (National Early Warning Score 2, Royal College of Physicians, 2017), with each parameter's official 0–3 sub-score mapped onto our 0–100 scale. This is an explicit *partial* alignment — NEWS2 also scores respiratory rate and consciousness level, not currently captured by the intake form, and this limitation is documented directly in the module's docstring rather than left implicit. Symptom and history weights remain internally defined, as NEWS2 does not cover patient-reported symptoms.

2.3 Point 3 — French/Arabic NLP depth

Partially addressed — flagging done, NER upgrade optional and not yet started

The limitation was already documented in `nlp_service.py` before this feedback arrived. What's new: two regression tests, `test_LIMITATION_negation_beyond_window_is_missed` and `test_LIMITATION_double_negation_not_handled`, make the current failure modes verifiable by running the suite rather than only stated in prose. The lightweight multilingual clinical NER exploration the feedback flagged as optional (*"if time allows"*) has not been started — listed as a candidate for the next iteration.

2.4 Point 4 — Testing and validation

Done

A 37-test automated suite now exists (`test_risk_scoring.py`, `test_negation.py`), covering score capping, NEWS2 threshold boundaries, concept deduplication, negation exclusion, and confidence computation. Full detail, including a real bug the suite caught, in Chapter 4.

2.5 Point 5 — Security/compliance basics

Not yet addressed

Not yet addressed, stated plainly rather than partially claimed. Three concrete next steps identified:

- **HTTPS:** not applicable to the current localhost development environment; will be enforced at deployment time via an Nginx reverse proxy with Let's Encrypt (or a managed platform with HTTPS by default).
- **Encryption at rest:** not implemented for sensitive fields (`symptoms_text`, `medical_history`). Evaluating column-level encryption (`sqlalchemy-utils EncryptedType`) as a first step.
- **Data retention policy:** not yet written — requires a product decision (retention window, applicable local regulation) before it can be documented meaningfully.

2.6 Point 6 — Synchronous pipeline

Documented tradeoff, not implemented — matches what was asked

The feedback explicitly framed this as "fine for now" and asked for awareness of the tradeoff, not immediate implementation. The admission pipeline (NLP + scoring) stays synchronous by design — it completes in well under a second and the frontend needs the result immediately. The AI Assistant's local LLM inference is the first component slow enough to justify async handling; the tradeoff and a concrete migration path (FastAPI `BackgroundTasks` + polling) are documented in `docs/async_assistant_note.md`, with the reasoning for not implementing it yet (single concurrent user per clinic) stated explicitly rather than left as an oversight.

2.7 AI Assistant — design guardrails

Done

Built before this feedback was received, and every requested guardrail was already in place by the time the feedback arrived — a useful, unplanned test of whether the design instincts held up. Full detail in Chapter 3.

Requested guardrail	Status
RAG over curated knowledge base, not a freeform chatbot	DONE
Non-substitutive disclaimer on every response	DONE
Every interaction logged (question, answer, sources)	DONE
Different access scopes for patients vs. doctors	DONE
Design note before implementation	PARTIAL (<i>written retroactively</i>)

New Feature: AI Assistant

CHAPTER

3.1 Two-mode architecture

The assistant routes deterministically between two modes based on whether a `patient_id` is supplied by the calling frontend context — never a freeform, unscoped LLM call in either case.

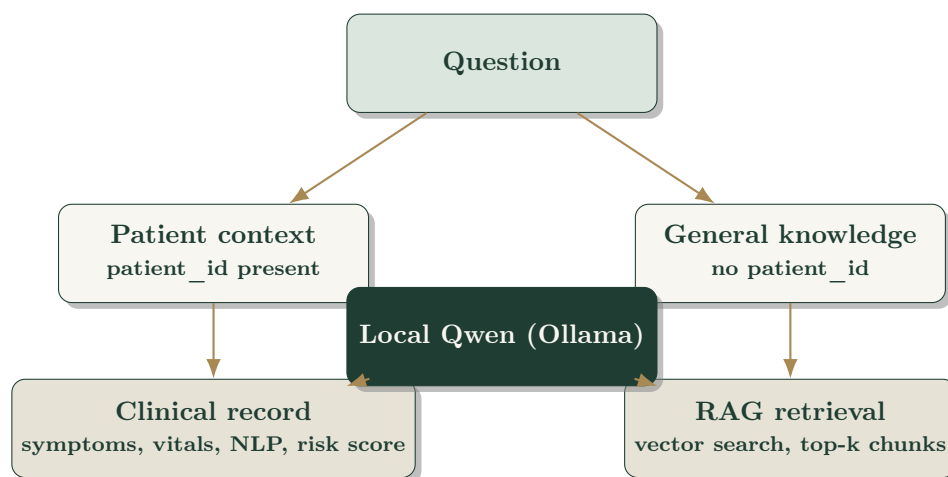


Figure 3.1: Deterministic routing — never a mixed or unscoped call

3.2 Guardrails implemented

- **Grounding, not memory** — the patient mode injects only structured data already persisted by the existing pipeline (`patient_intakes`, `extracted_entities`, `risk_assessments`); the general mode injects only retrieved document chunks, with mandatory source citation in the prompt.
- **Disclaimer** — every response carries the same non-substitutive disclaimer used elsewhere on the platform.
- **Audit log** — every exchange is persisted in `ai_conversations` (question, answer, mode, sources, user, patient), mirroring the existing AI-vs-clinician audit log.
- **Access control** — both modes are currently doctor/admin only, with the same clinic-isolation rule (`_can_access_patient`) used throughout the platform. A patient-facing general-question mode is explicitly deferred, not shipped by default, pending a separate risk review.

Testing & Validation

CHAPTER

4.1 Suite overview

File	Coverage	Tests
test_risk_scoring.py	Score capping, NEWS2 boundaries, concept dedup, negation exclusion, confidence	23
test_negation.py	EN/FR negation, clause breaks, documented limitations, field-boundary regression	14
Total		37

4.2 Case study — a real bug caught by writing tests

Found during manual QA, confirmed and fixed with regression tests

A submitted admission with `symptoms_text` ending without a period (e.g. *"No fever and no cough"*) caused `medical_history` to merge into the same sentence during NLP preprocessing (*"...no cough history of hypertension..."*). The negation window for *"no cough"* then incorrectly extended into the next field, causing `hypertension` and `heart attack` to be misclassified as negated and silently excluded from the risk score.

Fix: field concatenation now goes through `_build_nlp_text()`, which enforces an explicit sentence boundary between `symptoms_text` and `medical_history` regardless of the punctuation the user typed.

Before fix	After fix
hypertension and heart attack missing from Contributing Factors, score 100/100 without them counted	Both correctly present: hypertension (+5), heart attack (+15)

Covered by three new tests in `TestFieldConcatenationBoundary`: `test_missing_period_no_longer_leaks`, `test_empty_medical_history_handled`, `test_none_medical_history_handled`.

Model Confidence — Deep Dive

CHAPTER

5.1 The formula

Confidence is explicitly *not* a probability that the risk assessment is clinically correct — it reflects how much, and how reliable, the information behind the score is. Five weighted factors, capped at 95% so the tool never presents an assessment as a certainty:

Factor	Weight	What it measures
Vital-sign completeness	45%	How many of the 4 vitals were provided
Vital-sign validity	20%	Provided vitals within a plausible physiological range
NLP entity yield	20%	0 / 1 / 2+ clinical entities detected
Concept diversity	10%	How many distinct clinical concepts (not just mentions)
Negation handling	5%	Penalizes heavily-negated admissions slightly

5.2 Worked examples — real output, not illustrative

Each row below was produced by calling `compute_risk()` directly and recording the actual returned value.

Scenario	Level	Confidence
No data at all (empty admission)	LOW	5%
One vital only (temperature), one symptom (headache), no negation	LOW	51%
Three vitals (temperature, heart rate, blood pressure), one symptom (fatigue), no negation	LOW	74%
Full critical case: chest pain, shortness of breath, dizziness, hypertension, heart attack (5 entities, 2 negated), all 4 vitals provided and valid	CRITICAL	95% (capped)

Reading the 51% / 74% cases

Both examples above happen to fall in the *low* risk band by score, yet their confidence differs by 23 points — this is the intended behavior: confidence tracks *data quality*, independently of the risk level itself. A clinician sees not just "low risk" but "low risk, but only 51% confidence" — a signal that more information (more vitals, a fuller history) would sharpen the assessment, distinct from a "low risk, 74% confidence" case where more data was actually available.

Status Summary

CHAPTER

Area	Current state	Status
Model confidence	Five-factor computed formula, capped at 95%, verified with real worked examples	DONE
NEWS2 vital scoring	Temperature, heart rate, SpO2, systolic BP aligned; respiratory rate and consciousness not captured (documented)	DONE
Field concatenation bug	Found via manual QA, fixed with explicit sentence-boundary handling, 3 regression tests added	DONE
Test suite	37 tests, 37 passing	DONE
AI Assistant — general mode (RAG)	Working: retrieval + Qwen generation + source citation	DONE
AI Assistant — patient-context mode	Working: embedded on the risk result page, doctor/admin only	DONE
AI Assistant — audit logging	ai_conversations table capturing every exchange	DONE
Clinical notes	Doctor/admin write, patient + clinic staff read, unchanged and stable this iteration	DONE
French/Arabic NLP depth	Limitation documented and now test-verified; multilingual clinical NER upgrade not started (optional per feedback)	PARTIAL
Async handling	Admission pipeline stays sync by design; AI Assistant migration path documented, not implemented (justified: low concurrent load)	PARTIAL
HTTPS enforcement	Not applicable in current localhost dev environment; deployment plan documented (Nginx + Let's Encrypt)	OPEN
Encryption at rest	Not implemented for sensitive fields; column-level encryption identified as next step	OPEN
Data retention policy	Not yet written; requires a product decision first	OPEN

Live Verification — Screenshots

CHAPTER

Every number quoted in Chapters 4 and 5 was first verified by calling the risk engine directly in isolation. This chapter shows the same test suite and the same confidence values, this time captured live from the running application — closing the loop between the offline calculation and the actual production behavior.

7.1 Automated test suite — live run

```
(venv311) PS D:\Clinical_AI\smart-healthcare-backend> python -m pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- D:\Clinical_AI\smart-healthcare-backend\venv311\Scripts\python.exe
cachedir: .pytest_cache
rootdir: D:\Clinical_AI\smart-healthcare-backend
plugins: anyio-4.15.1
collected 37 items

tests/test_negation.py::TestNegationEnglish::test_simple_negation_detected PASSED [ 2%]
tests/test_negation.py::TestNegationEnglish::test_no_negation_when_absent PASSED [ 5%]
tests/test_negation.py::TestNegationEnglish::test_negation_stops_at_clause_break_but PASSED [ 8%]
tests/test_negation.py::TestNegationEnglish::test_negation_stops_at_comma PASSED [ 10%]
tests/test_negation.py::TestNegationFrench::test_pas_de_detected PASSED [ 13%]
tests/test_negation.py::TestNegationFrench::test_sans_detected PASSED [ 16%]
tests/test_negation.py::TestNegationFrench::test_negation_stops_at_mais PASSED [ 18%]
tests/test_negation.py::TestNegationKnownLimitations::test_LIMITATION_negation_beyond_window_is_missed PASSED [ 21%]
tests/test_negation.py::TestNegationKnownLimitations::test_LIMITATION_double_negation_not_handled PASSED [ 24%]
tests/test_negation.py::TestSentenceSplitting::test_splits_on_period PASSED [ 27%]
tests/test_negation.py::TestSentenceSplitting::test_splits_on_multiple_delimiters PASSED [ 29%]
tests/test_negation.py::TestFieldConcatenationBoundary::test_missing_period_no_longer_leaks_negation PASSED [ 32%]
tests/test_negation.py::TestFieldConcatenationBoundary::test_empty_medical_history_handled PASSED [ 35%]
tests/test_negation.py::TestFieldConcatenationBoundary::test_none_medical_history_handled PASSED [ 37%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_never_exceeds_100 PASSED [ 40%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_never_negative PASSED [ 43%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[0-low] PASSED [ 45%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[14-low] PASSED [ 48%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[15-medium] PASSED [ 51%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[39-medium] PASSED [ 54%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[40-high] PASSED [ 56%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[69-high] PASSED [ 59%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[70-critical] PASSED [ 62%]
tests/test_risk_scoring.py::TestScoreCapping::test_score_to_level_boundaries[100-critical] PASSED [ 64%]
tests/test_risk_scoring.py::TestDeduplication::test_same_concept_counted_once PASSED [ 67%]
tests/test_risk_scoring.py::TestNegationExclusion::test_negated_entity_excluded_from_score PASSED [ 70%]
tests/test_risk_scoring.py::TestNegationExclusion::test_mixed_negated_and_present PASSED [ 72%]
tests/test_risk_scoring.py::TestNegationExclusion::test_zero_weight_entity_excluded PASSED [ 75%]
tests/test_risk_scoring.py::TestNEWS2Vitals::test_normal_vitals_contribute_zero PASSED [ 78%]
tests/test_risk_scoring.py::TestNEWS2Vitals::test_critical_oxygen_saturation PASSED [ 81%]
tests/test_risk_scoring.py::TestNEWS2Vitals::test_low_systolic_bp_flagged PASSED [ 83%]
tests/test_risk_scoring.py::TestNEWS2Vitals::test_malformed_blood_pressure_ignored_gracefully PASSED [ 86%]
tests/test_risk_scoring.py::TestModelConfidence::test_confidence_baseline_with_no_data PASSED [ 88%]
tests/test_risk_scoring.py::TestModelConfidence::test_confidence_increases_with_data_completeness PASSED [ 91%]
tests/test_risk_scoring.py::TestModelConfidence::test_confidence_never_exceeds_095 PASSED [ 94%]
tests/test_risk_scoring.py::TestModelConfidence::test_invalid_vital_does_not_count_as_quality PASSED [ 97%]
tests/test_risk_scoring.py::TestModelConfidence::test_single_vs_multiple_entities_changes_nlp_score PASSED [ 100%]

===== 37 passed in 1.54s =====
(venv311) PS D:\Clinical_AI\smart-healthcare-backend>
```

Figure 7.1: `python -m pytest tests/ -v` — 37 of 37 tests passing on the actual project, including the updated `TestModelConfidence` suite (Chapter 5) and the field-concatenation regression tests (Chapter 4).

7.2 Confidence formula — live values match predictions exactly

The three confidence values below were predicted analytically in Chapter 5 before being reproduced through the actual patient intake form, NLP pipeline, and risk engine end to end. All three match the predicted value exactly.

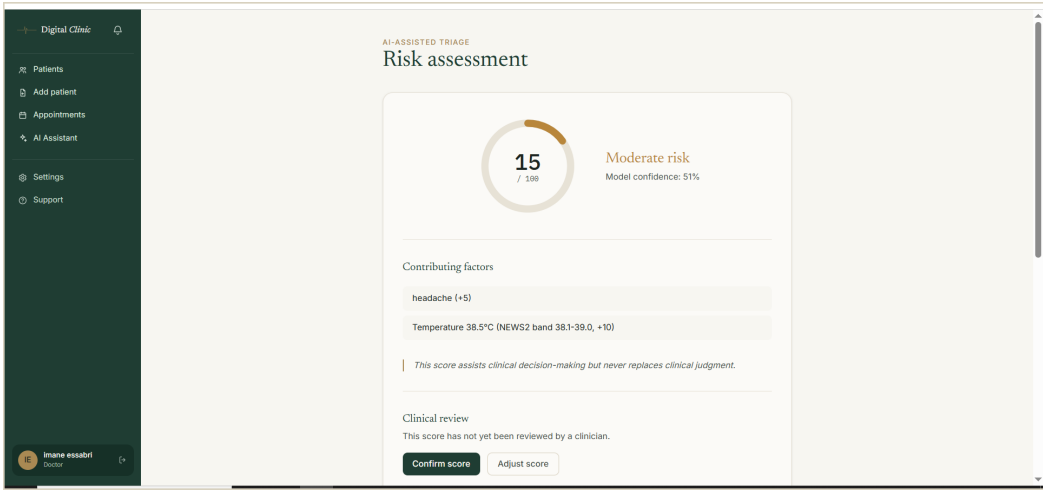


Figure 7.2: Single vital (temperature) and one symptom (headache), no negation — **51% confidence**, exactly as predicted.

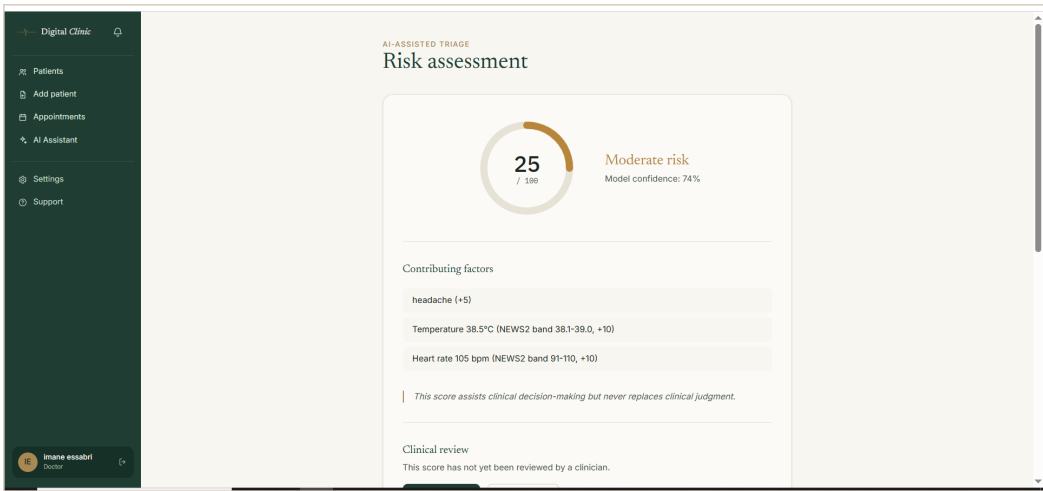


Figure 7.3: Two vitals (temperature, heart rate) and one symptom, no negation — **74% confidence**, exactly as predicted.

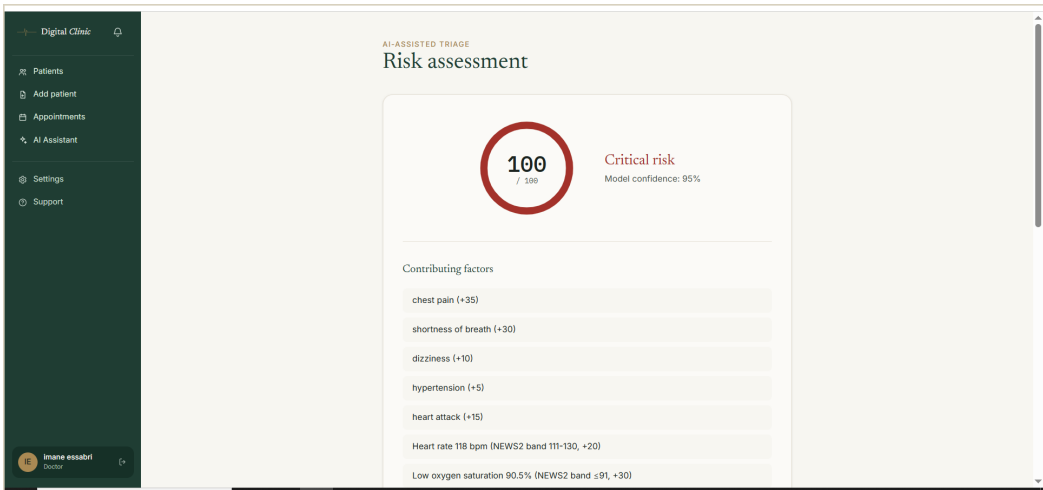


Figure 7.4: Full critical case — chest pain, shortness of breath, dizziness, hypertension, heart attack, all 4 vitals provided — **95% confidence (capped)**, matching the critical-case scenario in Chapter 4 where the field-concatenation bug was originally found and fixed.

7.3 AI Assistant — patient-context mode, live

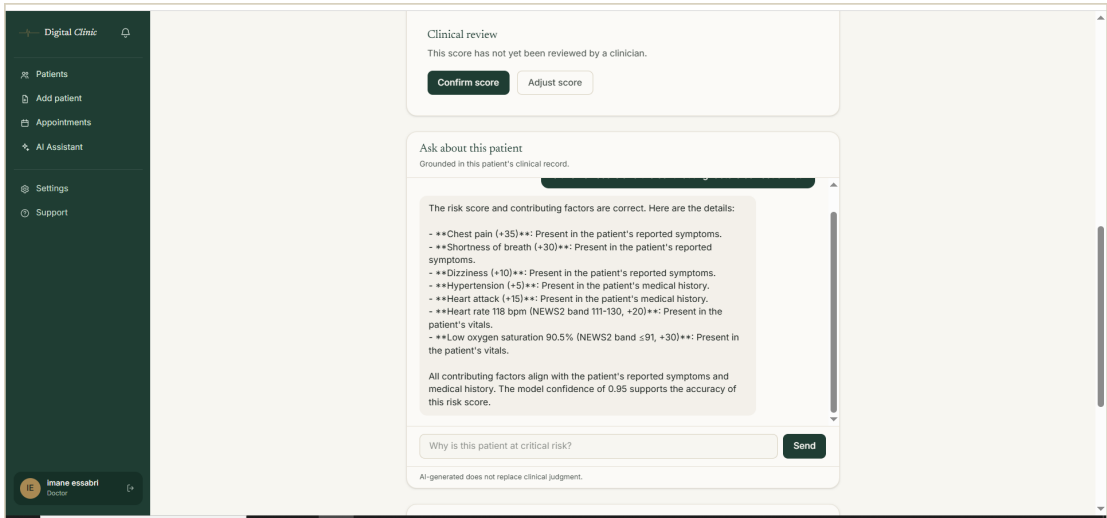


Figure 7.5: The AI Assistant, embedded on the same risk result page, correctly explains each contributing factor by cross-referencing the patient’s reported symptoms, medical history, and vitals — and cites the same 0.95 confidence value shown above, confirming the assistant reads from the same computed risk assessment rather than a separate or stale source.

Why this matters

The assistant’s answer is grounded entirely in structured data already computed by the risk engine — it does not recompute or reinterpret the score, it explains the one that was actually persisted. This is the patient-context guardrail from Chapter 3 demonstrated end to end, not just described.

7.4 AI Assistant — general mode (RAG), live

The general-knowledge mode described in Chapter 3 (*no patient_id* supplied) is shown below on the dedicated AI Assistant page. The question “*What is Hypertension (High Blood Pressure)*” is answered only from retrieved chunks of the clinic’s reference documents: the answer carries an inline citation ([Source 1: hypertension.md]), the retrieved files are listed beneath each message, and the non-substitutive disclaimer stays visible under the input box.

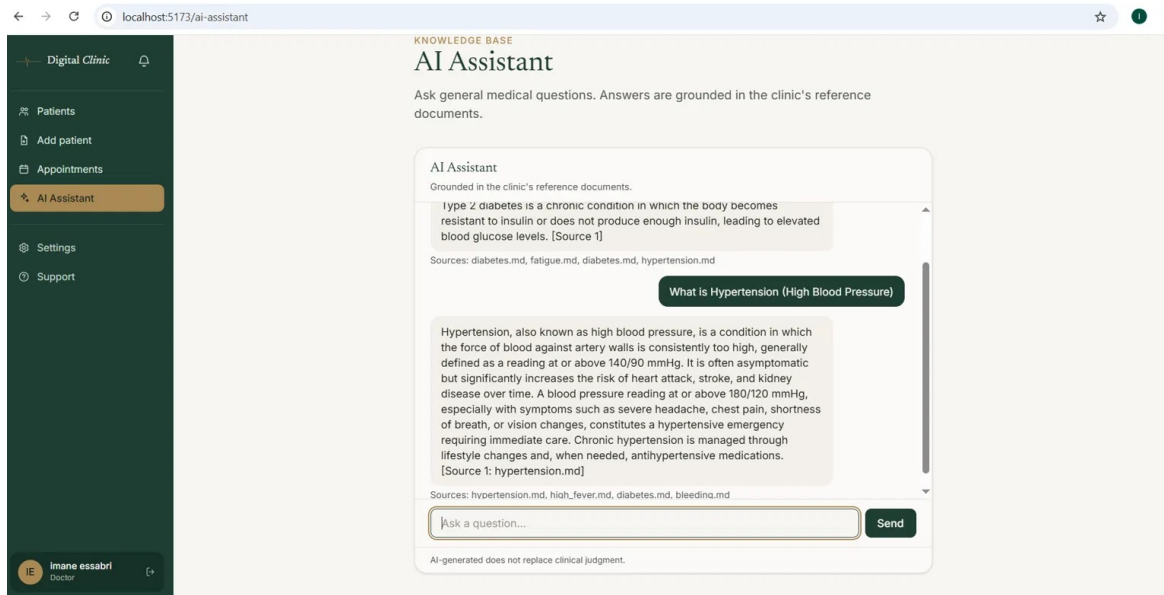


Figure 7.6: AI Assistant, general mode (RAG) — answer grounded in the clinic’s reference documents, with inline source citation, list of retrieved files, and the persistent non-substitutive disclaimer.

Why this matters

Unlike the patient-context mode (Section 3), this mode never reads a patient record. Every statement is traceable to a named source file, so the clinician can verify it against the clinic’s own documentation rather than trusting the model’s memory.

Conclusion

CHAPTER

This iteration turned six review points into either a shipped fix, a documented and justified trade-off, or an honestly flagged open item — no point was left ambiguous. The two most substantial changes, the NEWS2-aligned scoring and the computed confidence formula, directly strengthen the project’s core claim to be a genuine clinical decision support tool rather than a black box: every number the clinician sees is now traceable to a cited scale or a transparent, inspectable formula.

Writing the test suite paid for itself immediately — it surfaced a real scoring bug (Chapter 4) that would otherwise have silently under-counted risk for any admission where the symptom description didn’t end in a period, an easy thing for a real user to do.

The AI Assistant, built ahead of the review feedback that later validated its design, is now grounded, disclaimed, logged, and access-scoped — the same rigor now visible across the rest of the platform.

Clinical Decision Support

The intelligence added this iteration explains and grounds the system’s output more thoroughly than before. The clinical decision remains, in every case, the responsibility of the healthcare professional.

Prepared for the XelerondAI team — Smart Healthcare Platform, Iteration 2, 2026.